ShopCo Customer Analytics Report

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Management Summary

1. How reliable is our customer data?

About 13–14% of customers have no value for education, the "value for money" rating or electronics spend. The gaps are not random: education is missing for 43% of customers in the West (under 10% elsewhere), and customers without an electronics value are six years older on average. The gaps were filled in by multiple imputation. The correction is small (51.10 instead of 51.30 EUR average monthly electronics spend), so the data are a sound basis for decisions.

2. What does "satisfaction" mean?

Satisfaction is not a single scale but two separate dimensions. *Value Satisfaction* covers prices, value for money and promotions; *Experience Satisfaction* covers staff, store and checkout. The two are related to different outcomes (see question 4) and should be tracked separately.

3. Do regions shop differently, and are there spending segments?

Region is not related to the preferred shopping channel, so channel strategy does not need to vary by region. By spending, customers fall into three segments. Each spends about 100 EUR per month on groceries and electronics combined, but splits this budget differently (Table A).

Table A: Customer segments

Segment	Share	Annual spend	Main characteristic
Tech Enthusiasts	32%	541 EUR	mostly electronics
Household Shoppers	26%	655 EUR	mostly groceries; least satisfied with the store
Balanced Shoppers	42%	582 EUR	even split; most satisfied with the store

4. What drives spending and premium upgrades?

Annual spend is highest at around age 43 (644 EUR, compared with 541 EUR at 20 and 509 EUR at 70). Customers one standard deviation higher on Value Satisfaction spend about 60 EUR more per year, while Experience Satisfaction is not related to spending. Premium upgrades depend on other factors: education (graduates have almost four times the odds of customers with a high school education), grocery spend (22% higher odds per extra 10 EUR per month) and, most of all, Experience Satisfaction; price perception plays no role. Price perception is therefore linked to how much customers spend, and the store experience to whether they upgrade. The upgrade model identifies about three quarters of actual upgraders.

5. Did the retention program work?

Yes. Since customers were assigned at random, the difference between the groups can be attributed to the program. Enrolled customers leave at a rate about 46% lower than non-participants of the same age: after 10 months, 62% of enrolled customers were still with ShopCo, compared with 41% of non-participants. A separate analysis of churn in the first six months leads to the same conclusion.

1 Task 1: Data Quality and Missing Values

1.1 Missingness mechanism

Three variables have missing values: `education_level` (13.74%), `sat_value` (12.64%) and `spend_electronics` (13.92%); all other variables are complete. To check whether the values are Missing Completely at Random (MCAR), customers with and without a value were compared on every other observed variable, using Welch t-tests for numeric variables and Pearson chi-squared tests for categorical and binary variables (Table 2).

Table 2: Missingness compared with the other observed variables

Variable	Missing (%)	Significant association ($p < 0.05$)	MCAR plausible?
`education_level`	13.74	region ($p < 0.001$), channel ($p = 0.040$)	No
`sat_value`	12.64	none	Yes
`spend_electronics`	13.92	age ($p = 0.004$)	No

Missingness in `education_level` is not related to age ($p = 0.347$) but is strongly related to region: 43.2% of customers in the West have no value, compared with 4.8% in the North, 8.5% in the East and 9.8% in the South. MCAR is therefore rejected; the values are more likely Missing at Random (MAR) given region. The association with preferred channel ($p = 0.040$) is weak. Since 51 tests were carried out, two or three p-values below 0.05 are expected by chance, so this result is not given much weight.

For `sat_value`, no comparison is significant (the smallest p-value is 0.097, for `months_observed`), so MCAR is plausible.

Missingness in `spend_electronics` is related to age ($p = 0.004$): customers without a value are on average 52.2 years old, compared with 46.1 years for the others (Figure 1). MCAR is rejected, and MAR given age is assumed. These tests can reject MCAR, but they cannot rule out that missingness depends on the unobserved values themselves (MNAR), so the analysis below relies on the MAR assumption.

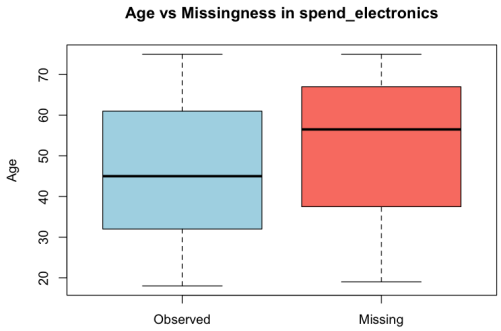


Figure 1: Age by missingness in `spend_electronics`

1.2 Multiple imputation

Multiple imputation was carried out with `mice` ($m = 20$, 10 iterations, seed 2026). Predictive mean matching (`pmm`) was used for the numeric variables `sat_value` and `spend_electronics`, and proportional odds logistic regression (`polr`) for the ordered factor `education_level`. PMM only draws values that occur in the observed data, so the imputed ratings stay on the 1–7 integer scale. All other variables, including region and age, were used as predictors; `customer_id` was excluded because it is only a row identifier.

The trace lines of the 20 chains are well mixed and show no trend across iterations (Figure 2), which indicates convergence. The imputed values of `spend_electronics` lie within the range of the observed values and have a similar spread (Figure 3).

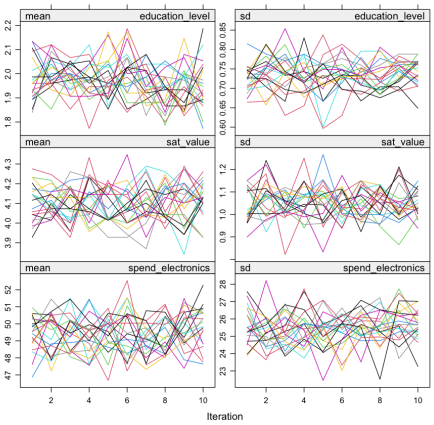


Figure 2: Convergence trace plot

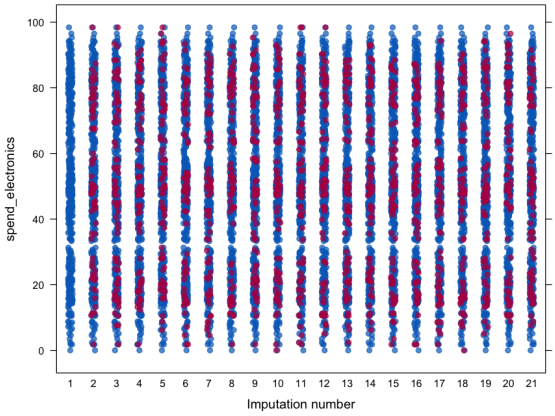


Figure 3: Imputed vs. observed values

1.3 Comparison of means

The complete-case mean of `spend_electronics` is 51.30 EUR. The mean pooled over the 20 imputed datasets with Rubin's rules is slightly lower, at 51.10 EUR (95% CI: 49.01 to 53.19 EUR).

The difference follows from the missingness mechanism. Values are missing more often for older customers, and in the observed data customers over 55 spend less on electronics (48.76 EUR at ages 56–65 and 47.66 EUR over 65) than those aged 36–55 (55.53 and 53.83 EUR). Dropping incomplete rows therefore under-represents older customers with lower spending, which pushes the complete-case mean up. The imputation uses age to fill in their values, so the pooled mean is lower. The difference (0.20 EUR) is small because the relationship between age and electronics spend is weak: the complete-case bias goes in the expected direction but is of little practical importance.

2 Task 2: What Does "Satisfaction" Mean?

All analyses in this task use the first completed dataset (imputation 1). This is acceptable because PCA, factor analysis and canonical correlation are used here to describe structure, not to test hypotheses with pooled standard errors, and only two of the variables involved were imputed (`sat_value` and `spend_electronics`). As a check, the PCA and the CCA were repeated on all 20 imputed datasets: the variance explained by the first two components ranged from 69.9% to 70.4%, and the first canonical correlation from 0.481 to 0.494. The choice of dataset therefore has no material effect.

2.1 Principal component analysis

PCA was carried out on the six standardised survey items. The first component (PC1) explains 40.4% of the variance and PC2 29.8%, together 70.2%. The scree plot (Figure 4) has a clear elbow after the second component, and only the first two eigenvalues are above 1 (2.42 and 1.79, compared with 0.56 for the third), so two components were kept.



Figure 4: Scree plot

Table 3: PCA loadings and varimax factor loadings

	PCA (unrotated)		Factor analysis	
Item	PC1	PC2	F1	F2
`sat_price`	0.401	−0.423	0.742	–
`sat_value`	0.470	−0.327	0.766	–
`sat_promotions`	0.404	−0.414	0.735	–
`sat_staff`	0.442	0.396	–	0.836
`sat_store`	0.337	0.456	–	0.653
`sat_checkout`	0.383	0.423	–	0.693
Variance explained	40.4%	29.8%	28.4%	27.2%

Factor loadings below 0.3 are not shown (–).

All six items load positively on PC1 (0.34 to 0.47), so PC1 is a general "overall satisfaction" index. PC2 is bipolar: the three price-related items load negatively (−0.33 to −0.42) and the three experience items positively (0.40 to 0.46), so PC2 contrasts satisfaction with prices against satisfaction with the store experience.

2.2 Factor analysis

In the PCA solution, PC1 mixes all six items and PC2 is a contrast between two groups of items, which makes the components hard to use as two separate measures. A factor model with two factors and varimax rotation separates the items into two groups, with no cross-loadings above 0.3 (Table 3):

- Factor 1, *Value Satisfaction*: `sat_price` (0.742), `sat_value` (0.766) and `sat_promotions` (0.735).
- Factor 2, *Experience Satisfaction*: `sat_staff` (0.836), `sat_store` (0.653) and `sat_checkout` (0.693).

The factors explain less variance than the components (28.4% and 27.2%, 55.6% in total). This is expected, because factor analysis only models the variance the items share and treats the rest as item-specific (uniquenesses from 0.28 for `sat_staff` to 0.57 for `sat_store`). The likelihood-ratio test does not reject the two-factor model ($\chi^2 = 3.56$, $df = 4$, $p = 0.469$), so two factors are enough. Both methods find the same two dimensions; the rotation simply lines up each factor with one group of items, which makes it easier to interpret. Regression scores for the two factors, oriented so that a higher score means higher satisfaction, were saved as `Sat_Value` and `Sat_Experience` for Tasks 4 and 5.

2.3 Canonical correlation analysis

The six survey items (set 1) were related to `spend_grocery`, `spend_electronics` and `annual_spend` (set 2). The first canonical correlation is moderate at 0.481, so the first pair of canonical variates share about 23% of their variance (0.481^2). Bartlett's test shows that the canonical correlations are jointly significant ($\chi^2 = 153.0$, $df = 18$, $p < 0.001$), but the second and third are not ($p = 0.384$), so only the first pair is interpreted.

Because the variables within each set are correlated, the variates were interpreted with structure correlations (the correlation of each variable with its own variate) rather than the weights. The satisfaction variate is defined by the value-related items, `sat_promotions` (0.875), `sat_price` (0.834) and `sat_value` (0.775), while the three experience items have correlations below 0.09. The spending variate is almost the same as `annual_spend` (0.980); `spend_grocery` (0.141) and `spend_electronics` (−0.100) play only a minor role. The standardised weights give the same picture.

Customers who are more satisfied with prices, value for money and promotions therefore tend to spend more in total, while satisfaction with staff, store and checkout is largely unrelated to spending. The association concerns the overall spending level, not the split between groceries and electronics. As CCA is correlational, it does not show that better price perception causes higher spending.

3 Task 3: Channels and Customer Segments

3.1 Channel preferences by region

Table 4 cross-tabulates region and preferred channel. Neither variable has missing values, so this part does not depend on the imputation. Pearson's chi-squared test gives $\chi^2 = 9.00$ ($df = 9$, $p = 0.438$); all expected counts are above 22, so the test is valid. There is no significant association between region and preferred channel.

Table 4: Preferred channel by region (row %), standardised residuals in brackets

Region	Catalog	In-store	Mobile App	Online
East	21.6 (−1.76)	26.1 (0.80)	28.1 (1.67)	24.2 (−0.61)
North	27.7 (0.27)	21.7 (−0.77)	22.9 (−0.13)	27.7 (0.60)
South	28.8 (0.55)	23.5 (−0.10)	24.2 (0.31)	23.5 (−0.76)
West	31.6 (1.13)	24.2 (0.10)	14.7 (−2.16)	29.5 (0.85)

A correspondence analysis (CA) was used to describe the pattern. The total inertia is very small (0.0165), in line with the weak association. The first dimension explains 84.7% of the inertia and the first two together 93.9% (Figure 5). The first dimension separates East and Mobile App from West, Catalog and, to a lesser extent, Online, while North and South lie close to the origin. This matches the row profiles: customers in the East use the Mobile App more (28.1%) and the Catalog less (21.6%) than average, whereas in the West only 14.7% prefer the Mobile App.

However, only one of the 16 cells (West, Mobile App) has a standardised residual beyond ±2, which is about what chance alone would produce. Together with the non-significant test, this suggests that the differences on the CA plot are sample-specific variation rather than real regional behaviour. Channel strategy does not need to differ by region on the basis of these data.

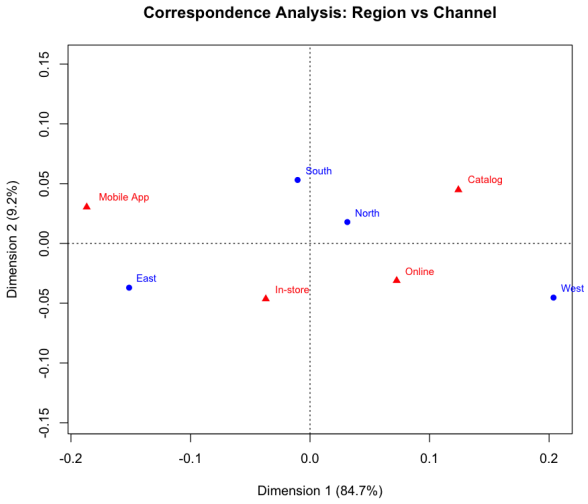


Figure 5: Correspondence analysis: region vs. preferred channel

3.2 Customer segmentation

Customers were clustered on `spend_grocery` and `spend_electronics`, using the same completed dataset as in Task 2. Both variables were standardised so that they carry equal weight in the distances. They are strongly negatively correlated ($r = -0.90$): customers who spend more on one category spend less on the other.

Hierarchical clustering (Euclidean distance, Ward's method) and k-means were both applied. The dendrogram (Figure 6) shows three main branches. The elbow plot (Figure 7) has a clear elbow at $k = 3$: the within-cluster sum of squares falls from 1090.0 ($k = 1$) to 379.3 ($k = 2$) and 114.7 ($k = 3$), and only slowly after that (97.1 for $k = 4$). The average silhouette width is also highest for $k = 3$ (0.649, compared with 0.561 for $k = 2$ and 0.508 for $k = 4$), so three segments were chosen. The two methods agree well: 540 of the 546 customers (98.9%) are placed in the same segment, and the other 6 lie on the borders between segments.

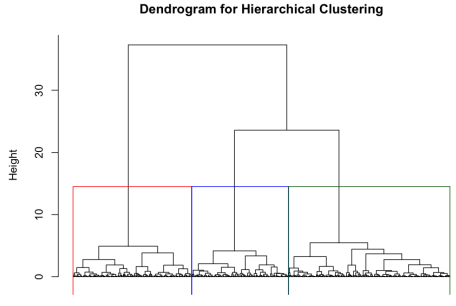


Figure 6: Dendrogram (Ward's method)

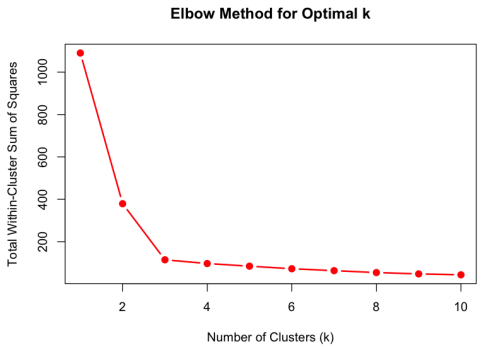


Figure 7: Elbow plot for k-means

3.3 Segment profiles

The k-means segments were profiled by size, spending, age and the two satisfaction scores (Table 5). Differences were tested with one-way ANOVA; the two clustering variables were not tested, since they differ by construction.

Table 5: Segment profiles (means) and one-way ANOVA p-values

	Tech	Household	Balanced	ANOVA p
Size (n)	175	143	228	–
Grocery spend (EUR/month)	20.94	81.42	51.01	–
Electronics spend (EUR/month)	79.38	19.47	49.59	–
Annual spend (EUR)	541.38	655.10	581.64	< 0.001
Age (years)	45.43	48.27	47.19	0.309
Value Satisfaction	−0.07	0.04	0.02	0.494
Experience Satisfaction	−0.04	−0.14	0.12	0.018

In every segment, the combined monthly spend on groceries and electronics is about 100 EUR, so the segments differ mainly in how they split this budget. This also explains the strong negative correlation between the two variables.

- *Tech Enthusiasts* ($n = 175$) spend most of their budget on electronics (79.38 EUR per month) and little on groceries (20.94 EUR). They have the lowest annual spend (541.38 EUR). Campaigns for this group should focus on electronics, such as product launches and accessories.
- *Household Shoppers* ($n = 143$) show the opposite pattern (81.42 EUR groceries, 19.47 EUR electronics). They have the highest annual spend (655.10 EUR) but the lowest Experience Satisfaction (−0.14). As the most valuable customers, they should be the first target for improvements to staff, store layout and checkout.
- *Balanced Shoppers* ($n = 228$), the largest group, split their budget evenly (51.01 and 49.59 EUR) and spend 581.64 EUR per year. They report the highest Experience Satisfaction (0.12), which makes cross-selling between the two categories a natural strategy.

The segments differ significantly in annual spend ($p < 0.001$) and Experience Satisfaction ($p = 0.018$), although the latter difference is modest (about a quarter of a standard deviation between Household and Balanced Shoppers). Age ($p = 0.309$) and Value Satisfaction ($p = 0.494$) do not differ: all segments are on average 45 to 48 years old and rate prices similarly. The segment labels are used in Task 4, with Tech Enthusiasts as the reference category.

4 Task 4: What Drives Annual Spend?

4.1 Linear and non-linear age effect

A simple linear regression of `annual_spend` on `age` was fitted first. Neither variable has missing values, so the original data were used. The model suggests that annual spend falls by 1.38 EUR per year of age ($p < 0.001$), but it explains only 3.0% of the variance. The residuals plotted against age (Figure 8) show a clear inverted U-shape in the loess curve: residuals are mostly negative for young and old customers and positive for middle-aged ones, so a linear age effect is not adequate.

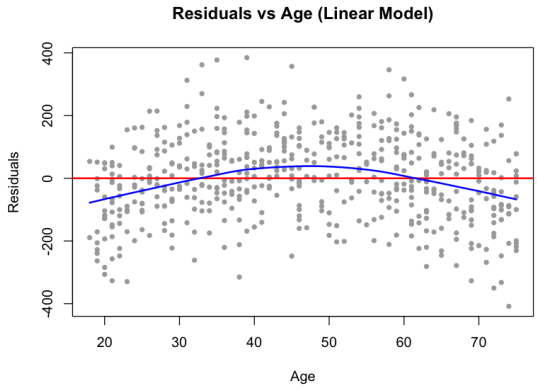


Figure 8: Residuals vs. age (linear model)

The linear term was therefore replaced by a quadratic one (age + age^2). The F-test comparing the two models is highly significant ($F = 78.22$, $p < 2.2 \times 10^{-16}$), the AIC drops from 6886.49 to 6815.02, and R^2 rises from 0.030 to 0.152, so the quadratic model fits clearly better. Predicted spending is highest at **43.3 years**. In euros, predicted annual spend rises from 541 EUR at age 20 to 611 EUR at 30 and 644 EUR at 43, and then falls to 618 EUR at 55 and 509 EUR at 70. A 43-year-old is thus expected to spend about 100 EUR more per year than a 20-year-old and about 135 EUR more than a 70-year-old. The negative slope of the linear model was misleading, because it averaged an increase up to the early forties with a decrease afterwards.

4.2 Adding satisfaction and segments

The quadratic age terms, the two satisfaction scores (Task 2) and the segments (Task 3, reference: Tech Enthusiasts) were combined in a multiple regression. The model was fitted on each of the 20 imputed datasets and pooled with Rubin's rules (Table 6).

Table 6: Multiple regression of annual spend (EUR), pooled over 20 imputations

Predictor	Estimate	95% CI	p-value
Intercept	243.78	[173.68, 313.88]	< 0.001
Age	16.52	[13.35, 19.69]	< 0.001
Age^2	−0.19	[−0.23, −0.16]	< 0.001
Value Satisfaction	66.91	[57.67, 76.16]	< 0.001
Experience Satisfaction	−4.14	[−13.39, 5.11]	0.380
Segment: Household (vs. Tech)	118.02	[96.28, 139.77]	< 0.001
Segment: Balanced (vs. Tech)	45.78	[26.36, 65.19]	< 0.001
$R^2 = 0.475$, adjusted $R^2 = 0.469$, $n = 546$			

The model explains 47.5% of the variance in annual spend (adjusted $R^2 = 0.469$). Joint F-tests show that age (both terms, $F = 86.29$, $p < 0.001$) and segment (both dummies, $F = 57.21$, $p < 0.001$) are significant. Holding the other variables constant, the results can be read in euros as follows.

Value Satisfaction is the only satisfaction dimension related to spending ($p < 0.001$). One unit higher on the score goes with 66.91 EUR more annual spend (95% CI: 57.67 to 76.16 EUR); since the score has a standard deviation of 0.89, a customer one standard deviation above average spends about 60 EUR more per year. Experience Satisfaction is not significant ($p = 0.380$) and its confidence interval (−13.39 to 5.11 EUR) includes zero, so satisfaction with staff, store and checkout is not related to how much customers spend. This agrees with the canonical correlation results in Task 2.

The segment differences are the largest effects in the model: Household Shoppers spend 118.02 EUR and Balanced Shoppers 45.78 EUR more per year than Tech Enthusiasts. The age effect remains after adding the other predictors, with predicted spending now highest at 42.7 years, very close to the 43.3 years in the age-only model.

Limitation. As allowed by the instructions, the satisfaction scores and segments come from one completed dataset (imputation 1) and were added unchanged to all 20 imputed datasets. Since `annual_spend` and `age` are complete, all predictors are then identical across the imputations, so the pooled results equal those of a single fit. The standard errors therefore do not reflect the uncertainty from imputing `sat_value` and `spend_electronics` and may be slightly too small. Given the low share of missing values and the stability shown in Task 2, this effect should be small.

5 Task 5: Who Upgrades to Premium?

5.1 Logistic regression and odds ratios

A logistic regression of `premium_upgrade` on `education_level`, `spend_grocery` and the two satisfaction scores was fitted on each of the 20 imputed datasets and pooled with Rubin's rules, so the uncertainty from imputing `education_level` is included in the standard errors. Education was dummy-coded with high school as the reference. As in Task 4, the satisfaction scores come from imputation 1, so their imputation uncertainty is not fully captured.

Table 7: Pooled odds ratios for premium upgrade (logistic regression)

Predictor	Odds ratio	95% CI	p-value
Education: bachelor (vs. high school)	2.031	[1.250, 3.300]	0.004
Education: graduate (vs. high school)	3.750	[2.103, 6.684]	< 0.001
Grocery spend (per 1 EUR)	1.020	[1.012, 1.029]	< 0.001
Grocery spend (per 10 EUR)	1.222	[1.126, 1.327]	< 0.001
Value Satisfaction	0.934	[0.753, 1.159]	0.535
Experience Satisfaction	2.772	[2.172, 3.538]	< 0.001

The odds of upgrading increase with education (Table 7). Compared with customers with a high school education, customers with a bachelor's degree have about twice the odds ($OR = 2.03$, 95% CI: 1.25 to 3.30) and graduates almost four times the odds ($OR = 3.75$, 95% CI: 2.10 to 6.68). Each extra euro of monthly grocery spend raises the odds by 2%, or by about 22% per 10 EUR ($OR = 1.22$, 95% CI: 1.13 to 1.33).

Experience Satisfaction has the largest effect of the four predictors: one unit higher on the score (roughly one standard deviation) multiplies the odds of upgrading by 2.77 (95% CI: 2.17 to 3.54). Value Satisfaction, which was the satisfaction dimension linked to annual spend in Task 4, has no effect on upgrading ($OR = 0.93$, $p = 0.535$; the CI includes 1). This difference makes business sense: customers who care about low prices may buy a lot, but a premium fee is more likely to be paid by customers who value the service and the store.

5.2 Probit comparison

The model was refitted with a probit link. The ratio $\beta_{logit}/\beta_{probit}$ lies between 1.65 and 1.70 for all coefficients, close to the usual rule of thumb of about 1.6, and the p-values are almost identical (for example 0.0043 vs. 0.0041 for bachelor, 0.535 vs. 0.542 for Value Satisfaction). The predicted probabilities, averaged over the 20 imputations, correlate at $r = 0.9998$; the largest difference for any customer is 0.013, and at a threshold of 0.5 no customer is classified differently. The choice between logit and probit therefore does not matter here.

5.3 Classification performance

The area under the ROC curve (Figure 9) is 0.767 (95% CI: 0.728 to 0.807), which indicates acceptable to good separation of upgraders and non-upgraders. Two thresholds were compared (Table 8). At the usual threshold of 0.5, accuracy is 69.6%, but only 57.0% of the actual upgraders (130 of 228) are found. The threshold that maximises sensitivity plus specificity (Youden index) is 0.378, close to the observed upgrade rate of 41.8%. At this threshold accuracy is practically unchanged (69.8%), while 75.0% of upgraders (171 of 228) are found, at the cost of more false positives (108 instead of 68).

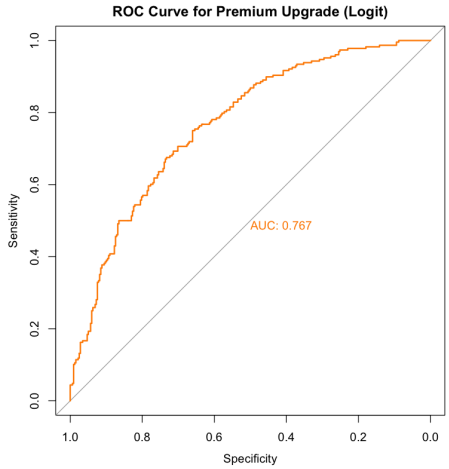


Figure 9: ROC curve (logit model)

Table 8: Confusion matrices at two thresholds

	Threshold 0.5	Threshold 0.378
True pos. / false neg.	130 / 98	171 / 57
True neg. / false pos.	250 / 68	210 / 108
Accuracy	69.6%	69.8%
Sensitivity	57.0%	75.0%
Specificity	78.6%	66.0%

The threshold of 0.378 is recommended. The two errors do not cost the same: sending an upgrade offer to a customer who does not take it (false positive) costs little, while missing a customer who would have upgraded (false negative) means lost subscription revenue. The lower threshold finds many more potential upgraders without lowering overall accuracy. Since the model was evaluated on the data it was fitted on, these figures are likely somewhat optimistic for new customers.

6 Task 6: Did the Retention Program Work?

6.1 Kaplan–Meier estimates

The time until a customer left (`months_observed`) was analysed with `churn_event` as the event indicator. None of the variables in this task has missing values, so the original data were used. Since customers were randomly assigned to the program, differences between the groups can be read as the effect of the program.

The Kaplan–Meier curve for enrolled customers (blue, dashed) lies above that for non-participants (red, solid) throughout, and the gap widens over time (Figure 10). After 6 months, an estimated 78.4% of enrolled customers are still with ShopCo, compared with 65.9% of non-participants; after 10 months the figures are 62.0% and 40.7%. Half of the non-participants have left after 8.3 months (median survival), while the median is not reached in the program group. The log-rank test confirms that the curves differ ($\chi^2 = 18.3$, $df = 1$, $p = 2 \times 10^{-5}$).

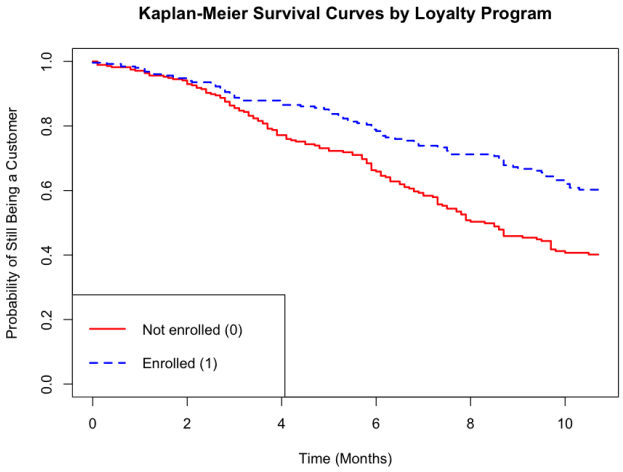


Figure 10: Kaplan–Meier curves by loyalty program participation

6.2 Cox proportional hazards model

A Cox model with `loyalty_program` and `age` was fitted (546 customers, 223 churn events). The Schoenfeld residual test (`cox.zph`) shows no violation of proportional hazards (program $p = 0.71$, age $p = 0.74$, global $p = 0.89$), so the hazard ratios can be taken as constant over time.

The program has a significant effect ($p < 0.001$). The hazard ratio is 0.541 (95% CI: 0.412 to 0.711): at any point in time, enrolled customers leave at a rate about **46% lower** than non-participants of the same age. Older customers are somewhat more likely to leave ($p < 0.001$): the churn rate rises by about 1.5% per year of age (HR = 1.015, 95% CI: 1.007 to 1.023), or about 16% per 10 years (HR = 1.158).

6.3 The role of censoring

Simply comparing the share of customers with `churn_event` = 1 (49.5% for non-participants vs. 31.4% for participants) would be a mistake. `churn_event` = 0 does not mean that a customer stayed, only that they had not left by the end of their observation period. Observation periods differ strongly: some customers were followed for the full 10.7 months, others for only a few months (98 customers were censored before month 6), and the median follow-up also differs between the groups (6.4 months without the program, 7.8 with it). A simple share treats a customer censored after two months like one who stayed the whole period, although we do not know whether the first would have left later. The share therefore depends on how long each customer happened to be observed and is not a churn rate over a fixed period. Kaplan–Meier and Cox models avoid this, because each customer contributes only the time they were actually observed.

6.4 Discrete-time comparison (6-month churn)

As a further check, a binary outcome "left within 6 months" was defined. The 98 customers censored before month 6 were excluded, because it is unknown whether they would have left before month 6. Of the remaining 448 customers, 36.0% of non-participants and 23.4% of participants left within 6 months.

Logit and probit models with `loyalty_program` and `age` both show a significant negative program effect (logit $p = 0.0027$, probit $p = 0.0025$). In the logit model, enrolled customers have about 47% lower odds of leaving within 6 months ($OR = 0.527$, 95% CI: 0.346 to 0.801), and older customers slightly higher odds ($OR = 1.017$ per year, $p = 0.009$). The ratio $\beta_{logit}/\beta_{probit}$ is 1.66 for the program and 1.65 for age, and the two models lead to the same conclusions.

These results agree with the Cox model: the program effect is of similar size (HR = 0.54 vs. OR = 0.53), and so is the age effect (1.015 vs. 1.017 per year). The hazard ratio and the odds ratio measure different things, however, so their similar values should not be read as the same effect. The Cox model is preferred, because it uses all 546 customers and their full observation time, whereas the discrete-time approach drops the 98 customers censored early and ignores when within the 6 months a customer left. Both approaches give strong evidence that the retention program reduces churn.